the lens of the binding problem. *Advances in Neural Information Processing Systems*, 37.

Stefan Depeweg, Contantin A Rothkopf, and Frank Jäkel. 2024. Solving bongard problems with a visual language and pragmatic constraints. *Cognitive Science*, 48(5):e13432.

Idan Fine and Omri Abend. 2025. A fine-grained analysis of multi-object representation in clip encoders. *arXiv preprint arXiv:2502.19842*. Submitted to ICLR 2025.

Harry E Foundalis. 2006. Phaeaco: A cognitive architecture inspired by bongard's problems. *Doctoral Dissertation, Indiana University*.

Dedre Gentner and Keith J Holyoak. 1997. Reasoning and learning by analogy. *American Psychologist*, 52(1):32–34.

Dedre Gentner and Christian Hoyos. 2017. Analogy and abstraction. *Topics in Cognitive Science*, 9(3):672–693.

James Gips. 1975. Shape grammars and their uses: Artificial perception, shape generation and computer aesthetics. *PhD thesis, Stanford University*.

Google DeepMind. 2024. Gemini 2.0 flash: Technical report. https://ai.google.dev/gemini-api/docs. Accessed October 2025.

Tetiana Grinberg, Dmitry Grinberg, Grigory Shcherbanskiy, and Lev Cherbanski. 2023. Bongard architecture: Towards scalable and transparent machine reasoning. *AAAI Spring Symposium Series EDGeS Submission*.

Daya Guo and 1 others. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Douglas R Hofstadter. 1979. *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books.

Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. 2023. Sugarcrepe: Fixing hackable benchmarks for vision-language compositionality. In *Advances in Neural Information Processing Systems*, volume 36, pages 38957–38977.

Yushi Hu, Benlin Liu, Jungo Kasai, Yizhong Wang, Mari Ostendorf, Ranjay Krishna, and Noah A Smith. 2023. Tifa: Accurate and interpretable text-to-image faithfulness evaluation with question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20406–20417.

Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling up visual and vision-language representation learning with noisy text supervision. In *International Conference on Machine Learning*, pages 4904–4916. PMLR.

Huaizu Jiang, Xiaojian Ma, Weili Nie, Zhiding Yu, Yuke Zhu, and Anima Anandkumar. 2022. Bongard-hoi: Benchmarking few-shot visual reasoning for human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19735–19745.

Cheongwoong Kang and Jaesik Choi. 2023. Impact of co-occurrence on factual knowledge of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 7770–7791.

Haoqiang Kang, Enna Sachdeva, Piyush Gupta, Sangjae Bae, and Kwonjoon Lee. 2025. Gflowvlm: Enhancing multi-step reasoning in vision-language models with generative flow networks. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 3815–3825.

Terry Knight and George Stiny. 2015. Making grammars: From computing with shapes to computing with things. *Design Studies*, 41:8–28.

Peter Kulits, Haiwen Feng, Weiyang Liu, Victoria Abrevaya, and Michael J Black. 2024. Re-thinking inverse graphics with large language models. *arXiv preprint arXiv:2404.15228*.

Tejas D Kulkarni, William F Whitney, Pushmeet Kohli, and Josh Tenenbaum. 2015. Deep convolutional inverse graphics network. In *Advances in Neural Information Processing Systems*, volume 28.

Quang-Hung Le, Long Hoang Dang, Ngan Hoang Le, Truyen Tran, and Thao Minh Le. 2025. Progressive multi-granular alignments for grounded reasoning in large vision-language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 4473–4481.

Haoxin Li, Bohao Liu, Hanyang Zhang, and Heng Huang. 2025. Enhancing vision-language compositional understanding via synthetic perturbations. *arXiv preprint arXiv:2503.10825*. CVPR 2025.

Zixian Ma, Jerry Hong, Mustafa Omer Gul, Mona Gandhi, Irena Gao, and Ranjay Krishna. 2023. Crepe: Can vision-language foundation models reason compositionally? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10910–10921.

Microsoft Research. 2024. Introducing phi-4: Microsoft's newest small language model specializing in complex reasoning. https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/introducing-phi-4.

Weili Nie, Zhiding Yu, Lei Mao, Ankit B Patel, Yuke Zhu, and Anima Anandkumar. 2020. Bongard-logo: A new benchmark for human-level concept learning and reasoning. In *Advances in Neural Information Processing Systems*, volume 33, pages 16468–16480.

Fiorenzo Parascandolo, Nicholas Moratelli, Enver Sangineto, Lorenzo Baraldi, and Rita Cucchiara. 2025. Causal graphical models for vision-language compositional understanding. In *International Conference on Learning Representations (ICLR)*.

Wujian Peng, Sicheng Xie, Zuyao You, Shiyi Lan, and Zuxuan Wu. 2024. Synthesize, diagnose, and optimize: Towards fine-grained vision-language understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26161–26171.

Yuxuan Qiao, Haodong Duan, Xinyu Fang, Junming Yang, Lin Chen, Songyang Zhang, Jiaqi Wang, Dahua Lin, and Kai Chen. 2024. Prism: A framework for decoupling and assessing the capabilities of vlms. In *Advances in Neural Information Processing Systems*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, and 1 others. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PMLR.

Rahul Ramachandran, Ali Garjani, Roman Bachmann, Andrei Atanov, Oğuzhan Fatih Kar, and Amir Zamir. 2025. How well does gpt-4o understand vision? evaluating multimodal foundation models on standard computer vision tasks. *arXiv preprint arXiv:2507.01955*.

Ruizhuo Song and Beiming Yuan. 2024. Solving the clustering reasoning problems by modeling a deep-learning-based probabilistic model. *arXiv preprint arXiv:2403.03173*.

George Stiny and James Gips. 1972. Shape grammars and the generative specification of painting and sculpture. In *Information Processing 71: Proceedings of the IFIP Congress 1971*, volume 2, pages 1460–1465. North-Holland.

Mohit Vaishnav and Tanel Tammet. 2025. A cognitive paradigm approach to probe the perception-reasoning interface in vlms. *arXiv preprint arXiv:2501.13620*.

Rujie Wu, Xiaojian Ma, Zhenliang Zhang, Wei Wang, Qing Li, Song-Chun Zhu, and Yizhou Wang. 2024. Bongard-openworld: Few-shot reasoning for free-form visual concepts in the real world. In *International Conference on Learning Representations (ICLR)*.

Antje Wüst, Sven Magg, and Stefan Wermter. 2024. Bongard in wonderland: Visual puzzles that still make ai models hallucinate. *arXiv preprint arXiv:2410.19546*.

Yichang Xu, Gaowen Liu, Ramana Rao Kompella, Sihao Hu, Tiansheng Huang, Fatih Ilhan, Selim Furkan Tekin, Zachary Yahn, and Ling Liu. 2025. Language-vision planner and executor for text-to-visual reasoning. *arXiv preprint arXiv:2506.07778*.

An Yang and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Ilker Yildirim, Mario Belledonne, Winrich Freiwald, and Josh Tenenbaum. 2020. Efficient inverse graphics in biological face processing. *Science Advances*, 6(10):eaax5979.

Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. 2022. When and why vision-language models behave like bags-of-words, and what to do about it? *arXiv preprint arXiv:2210.01936*.

Appendix

A Example Bongard-LOGO Instance

To illustrate the different symbolic views used in our experiments, we present a single Bongard-LOGO problem from the Basic (BD) category. The underlying concept combines an unbalanced trapezoidright triangle configuration with an uneven band of four arcs.

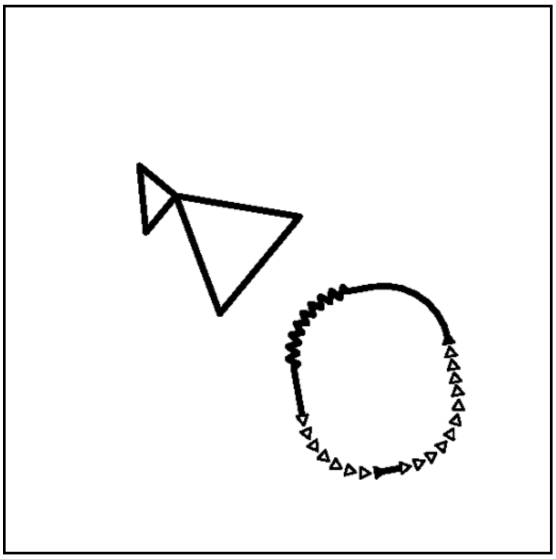
A.1 Concept and Natural-Language Description

This problem is labeled with the high-level Basic concept:

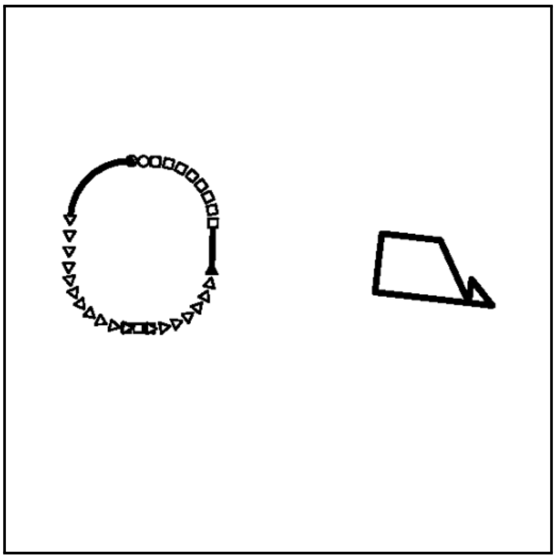
- **Concept UI:** "unbalanced trapezoid right_triangle AND uneven band four arcs"

A.2 Visual Layout

Figure A.1 shows the one positive and one negative example image.



(a) Negative example



(b) Positive example

Figure A.1: Example Basic (BD) Bongard-LOGO problem. Both the images conforming to the "unbalanced trapezoid right_triangle AND uneven band four arcs" concept.

A.3 Procedural Action Programs

Table A.1 shows the LOGO-style action programs for one negative and one positive example. Each token encodes both a primitive type and quantized geometric parameters (e.g., length, radius, angle).

A.4 Natural-Language Action Descriptions

For the C–G (Action Description) condition, each program is rendered as a stepwise English procedure. Below we show the corresponding descriptions for the same negative and positive examples.